(18)

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY,

LONERE – RAIGAD -402 103

Semester Examination – December – 2019

Sem.: - VII

Branch: Chemical/ Petrochemical

Subject with Subject Code: - Transport Phenomena

Marks: 60

Date: - 17/12/2019

Time: - 3Hr.

Instructions to the Students:

1. Each question carries 12 marks. Figure to right indicates marks.
2. Attempt any five questions of the following.
3. Illustrate your answers with neat sketches, diagram etc., wherever necessary.
4. If some part or parameter is noticed to be missing, you may appropriately assume it and should mention it clearly.

Q.1 **a)** Derive an expression for shape of the surface of a rotating liquid. **[6M]**

b) Water is flowing down a vertical wall having width with Re=10. Calculate, the film thickness in inches

DATA: Density of water=998 kg/m^3

Dynamic viscosity of water= 1 cp **[6M]**

Q.2 **a)** Derive an equation of motion. **[6M]**

b) Compute the steady state momentum flux in lbf/ft^2 when the lower plate velocity V is 3 ft/sec in the positive x direction. The plate separation Y is 0.002 ft and the fluid viscosity is 0.8 cp. **[6M]**

Q.3 **a)** Calculate the steady state mass flux j_{Ay} of helium for the system at 500K. The partial pressure of helium is 1 atm at y=0 and zero at the upper surface of the plate. The thickness Y of the pyrex plate is 10^{-2} mm, and its density (ρ^B) is 2.6 g/cm^3. The solubility and diffusivity of helium in pyrex are reported as 0.0084 volumes of gaseous helium per volume of gas, and $D_{AB} = 0.2*10^{-7}$ cm^2/s respectively. **[6M]**

b) Explain Shell energy balance equation and boundary conditions associated with it. **[6M]**

Q.4 For an electric wire of circular cross section with radius R, prove that

$$T-T_0= (S_cR^2/4k)[1-(r/R)^2]$$ **[12M]**

Q.5 Derive an expression for concentration profile of steady state diffusion of liq. A through stagnant gas film B. **[12M]**

Q.6 **a)** Glycerine ($CH_2OH.CHOH.CH_2OH$) at 26.5°C is flowing through a horizontal tube 1 ft long and with 0.1 in inside diameter. For a pressure drop of 40 psi , the volume flow rate w/ρ is 0.00398 ft_i^3/min . the density of glycerine at 26.5°C is 1.261 g/cm^3. From the flow data , find the viscosity of glycerine in centipoises and in Pa.s. **[6M]**

b) Write a note on temperature and pressure dependence of diffusivities. **[6M]**

—X—